



ATLAS SIGNAL DESK

Context-Aware Financial Intelligence

A conversational AI research assistant for financial analysis · *Built by*
Fahmida Fathima

The Problem

Market Data

Current information scattered across dozens of disconnected sources

Documents

Extracting signals from dense, lengthy financial reports

Context

Traditional assistants forget your role, interests, and watchlist

Atlas brings these workflows into one conversational interface.



Meet Atlas Signal Desk

Communicate naturally through Telegram. Ask financial questions without navigating complex dashboards.



Live Market Data

Real-time prices, market cap & sector intelligence



Voice Queries

Send spoken questions via Telegram voice messages



Chart Intelligence

Upload & interpret financial charts visually



Persistent Memory

Remembers your analyst profile across sessions



PDF Analysis

Upload financial reports for structured extraction



Natural Language

Context-aware responses tailored to your research focus

How Atlas Works

User Input



Atlas Orchestration

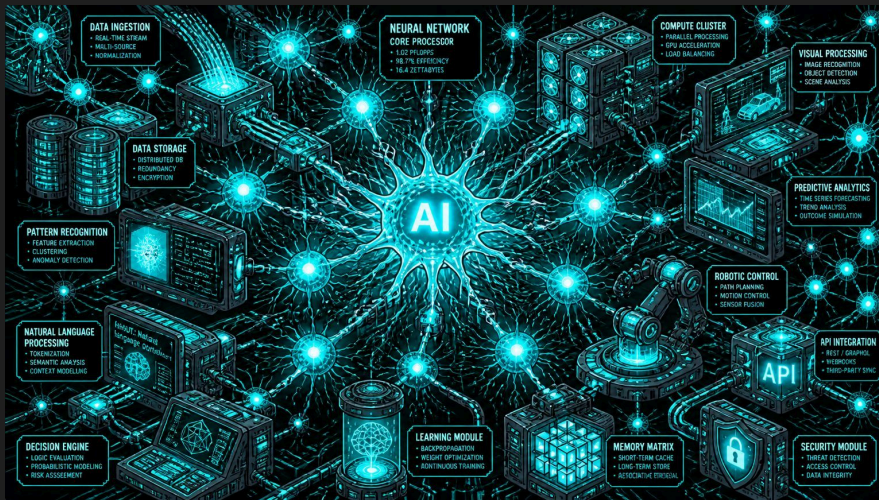
Gemini Reasoning

Python Tools

Gemini selects tools dynamically based on user intent and available data.

Agentic Tool Use

Atlas **reasons** about which tools to invoke — not just a chatbot.



Market Tool

Stock price, market cap, sector & industry via yfinance

News Tool

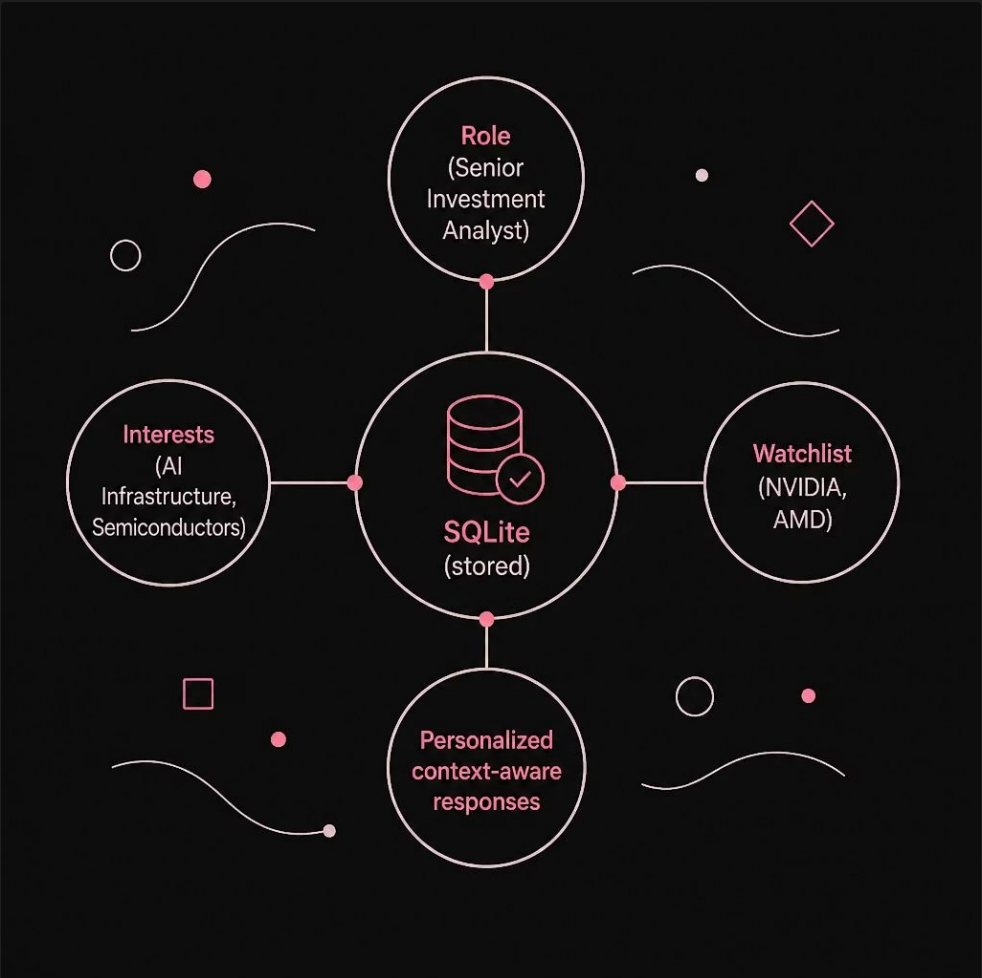
Recent company developments for research context

Memory Tool

Reads & updates analyst profile — role, interests, watchlist

Persistent Analyst Memory

Atlas remembers what matters — stored in SQLite for deeply personalized conversations.



Example Analyst Profile

Role Senior Investment Analyst	Interests AI Infrastructure · Semiconductors
Watchlist NVIDIA · AMD	

Future conversations adapt to your professional context and research focus.



Multimodal Financial Intelligence

One assistant. Multiple input types.

Text

Ask financial questions naturally

Voice

Telegram voice messages decoded instantly

PDF

Financial reports structured on upload

Image

Charts interpreted with visual AI

The ATLAS SIGNAL Framework

Every response follows a structured analyst-oriented format.



Watch Next

What to monitor going forward



Analyst Angle

Interpreted through your role & interests



Why It Matters

Financial significance explained



What Happened

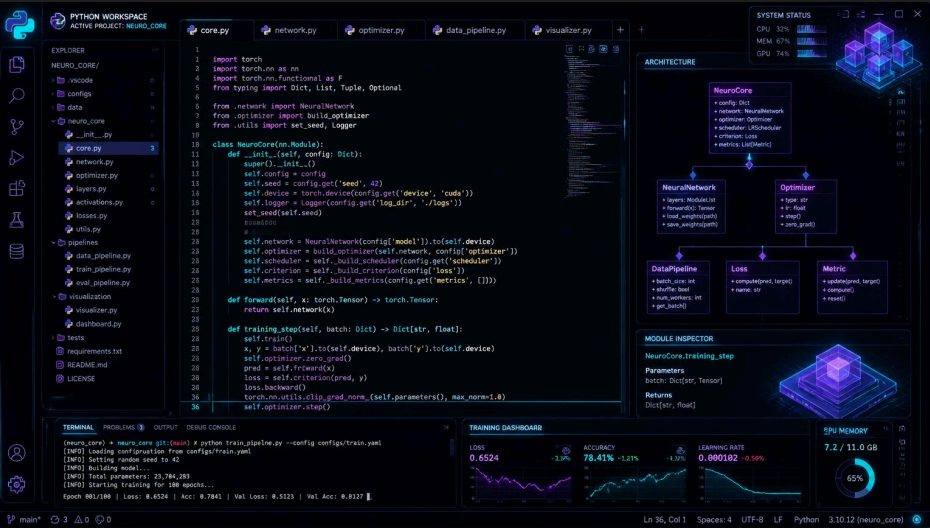
The key event or market information



"What's the latest signal on NVIDIA?" → Market data + news → structured ATLAS SIGNAL response

Engineering & Technology

Tech Stack



 AI Google Gemini · GenAI SDK	 Backend Python · python-telegram-bot
 Market Data yfinance	 Documents pypdf
 Memory SQLite · python-dotenv	

Modular Python architecture with tool-based components.

Live Demo & Impact

Try Atlas Now → https://t.me/AtlasSignalDeskAI_bot

01

Start Atlas

Ask a live market question

02

Tool Reasoning

Watch market data & tool selection in action

03

Memory Context

Ask a follow-up using remembered analyst profile

04

Voice & Multimodal

Send voice query · upload PDF & chart

05

ATLAS SIGNAL

View the structured analyst-grade response

"Atlas turns fragmented financial research into a conversational intelligence workflow."

Built by **Fahmida Fathima**